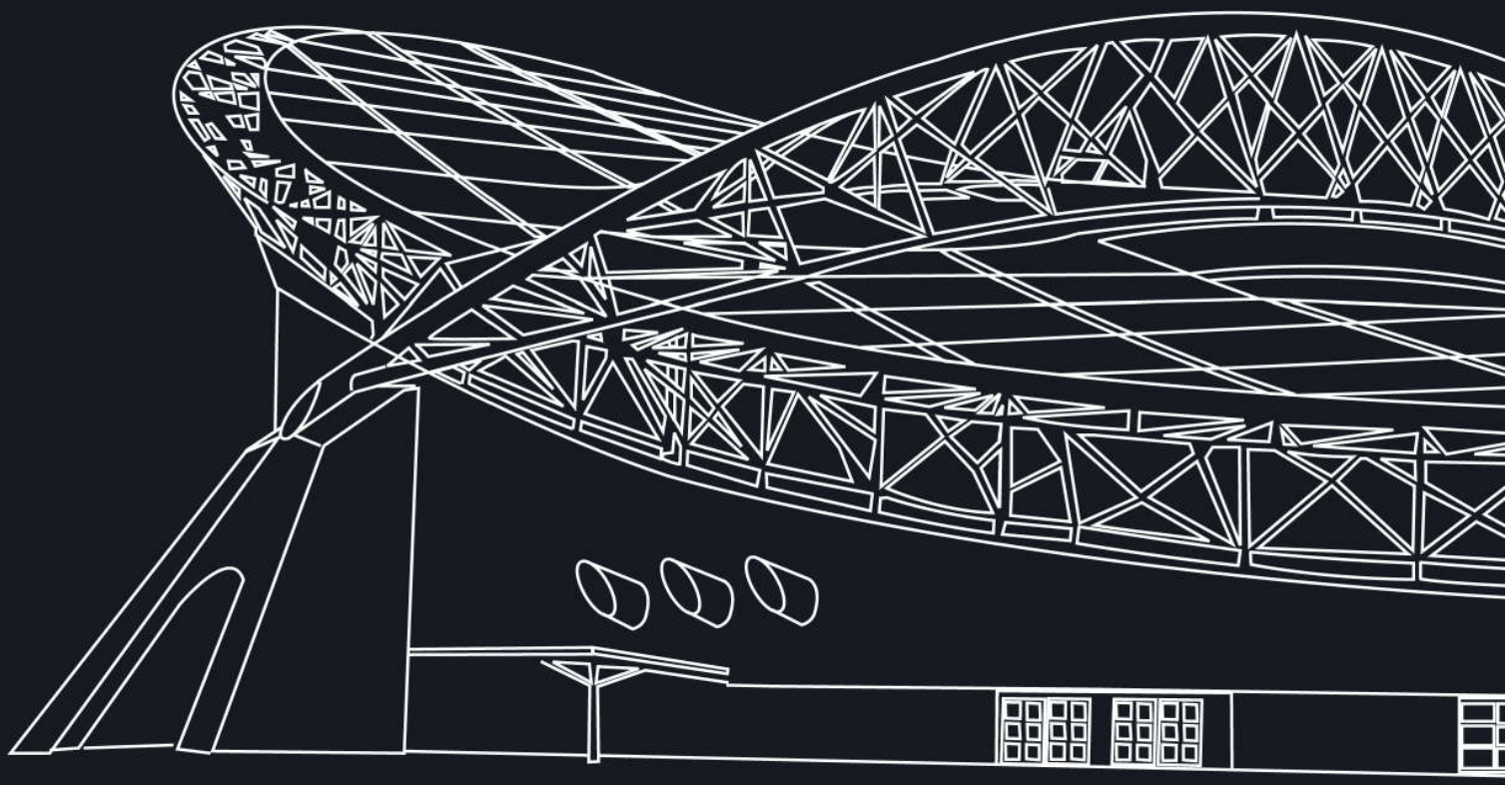
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Tata Structura

Rectangular, Square and Circular Steel Hollow Sections



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Profile

Located at Jamshedpur, Tata Steel Ltd. was established in 1907 and was the first Integrated Steel Plant in India. It commenced its operations in 1911, with a capacity of 1,00,000 tonnes per annum of ingots and expanded to a million tonnes per annum of saleable steel by the mid fifties.

Presently it has an annual capacity of 6 million tonnes at its Jamshedpur Plant.
The Plant, equipped with the latest steel making facilities such as Basic Oxygen Furnaces, Vacuum Degassing and Continuous Casting Units, presents the very epitome of technological advancement. A whole range of steels addressing the needs of the Construction/Engineering and automotive sector, testify to Tata Steel's capability. These products have found acceptance not only in the domestic market but all over the world.

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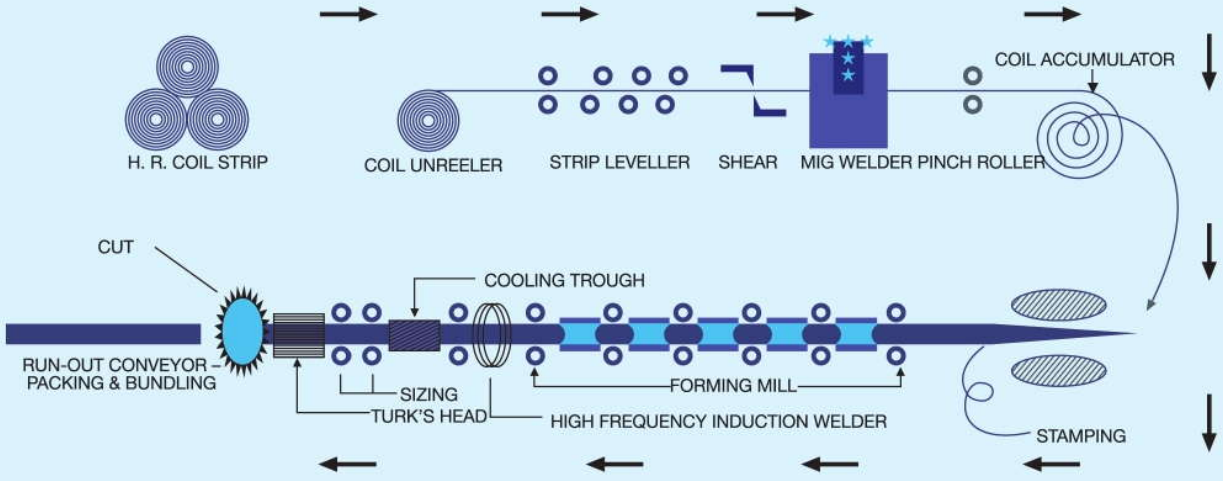
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The Tubes Division of Tata Steel came into being in 1985, after the merger of the erstwhile Indian Tube Company with Tata Steel. The Division manufactures commercial and precision tubes at its two plants, namely, the Standard Tubes Plant and the Precision Tubes Plant. The Tubes Strategic Business Unit of Tata Steel today is the largest manufacturer of a variety of steel tubes in India with an annual production capacity of around 4,00,000 tonnes and major expansion plans on the anvil to manufacture higher sizes. The SBU has its plants situated in Jamshedpur and has a network of sales offices across the country with the marketing headquarters in Kolkata to provide better customer service.

Tata Structura – Steel Hollow Section production process begins with the slitting of wide Steel Hot Rolled Coils to the width required for the production of customer specific section dimensions. This slit coil is then fed through the mill and is shaped into a round tube by a series of forming rolls. The converging strip edges of this round tube are squeezed together and welded using high frequency induction welding. Immediately after this external excess weld bead is removed. The tube then passes through a set of shaping rolls where it is shaped in to its final form either Square, Rectangle or Circular and is exactly sized to the required dimensions. Cutting to the required length and bundling are the final steps in this fully automatic manufacturing process.

Presently Tata Structura Rectangular & Square Steel Hollow Sections can be supplied to IS:4923 upto a maximum of 250 x 250 mm for square sections and 300 x 200 mm for rectangular sections. Circular Sections can be supplied to IS:1161 upto a maximum NB of 300 mm. However, with expansion plans, Tata Steel will be able to augment the size range for all sections.

The Process



Quality Control

The quality of our product is controlled during the manufacturing process. It starts with slitting the strip edges, continues with speed, temperature control during the high frequency induction welding and is followed by non destructive eddy current testing directly after welding. Off-line drift and flattening tests were conducted. This is all within our ISO 9000 quality management system.

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Advantages of Tata Structura Steel Hollow Sections

Tata Structura Steel Hollow Sections manufactured by Tata Steel command several techno-economic advantages over Conventional Steel Sections.

1. The raw material used for manufacturing these steel hollow sections comes from Tata Steel's state-of-the-art Hot Strip Mill, which manufactures Hot Rolled Coils, comparable to the best in the world.
2. The excellent distribution of material around the axis of these steel hollow sections, allows for remarkable strength qualities and thus offers decisive advantages as regards application technology.
3. Due to their high torsional rigidity and compressive strength, they behave more efficiently than Conventional Steel Sections.
4. Their higher strength to weight ratio could result in upto 30% savings in steel.
5. The smooth, uniform profile of these sections minimises dust accumulation and facilitates easy, on-site fabrication.
6. Tata Structura Steel Hollow Sections bring about a significant enhancement in the aesthetic appeal of structures.

Internal Corrosion - a case study



Sample 1



Sample 2

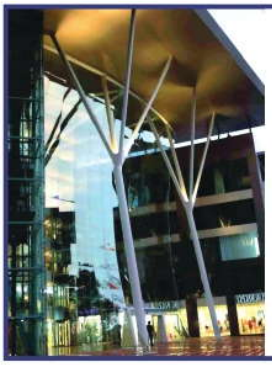
Two of the original 'Tubewrights' erected in 1954 at Stamford Bridge, Chelsea, were replaced in 1975, taken down and the used sections were cut and despatched to the Corby Works of British Steel for examination.

Sample No. 1 of the 139.7 mm o.d. CHS was cut lengthwise to expose the internal surfaces for examination. Little evidence of internal corrosion was found other than a discolouration of the surface caused by the oxygen and moisture in the entrapped air, much of the original mill scale was still visible. A light rust in the centre of the sample developed after the tube was cut open for examination.

Sample No. 2 of the 139.7 mm o.d. CHS incorporating an intermediate flanged joint was examined and the condition of the internal flange face, which had been enclosed and thus hermetically sealed by welding to the CHS, was still comparatively bright, with the oiginal marking-off lines clearly visible.

The above report is from British Steel Publication No. TD 347/10E/91 titled - CORROSION THE CASE for STRUCTURAL HOLLOW SECTIONS.

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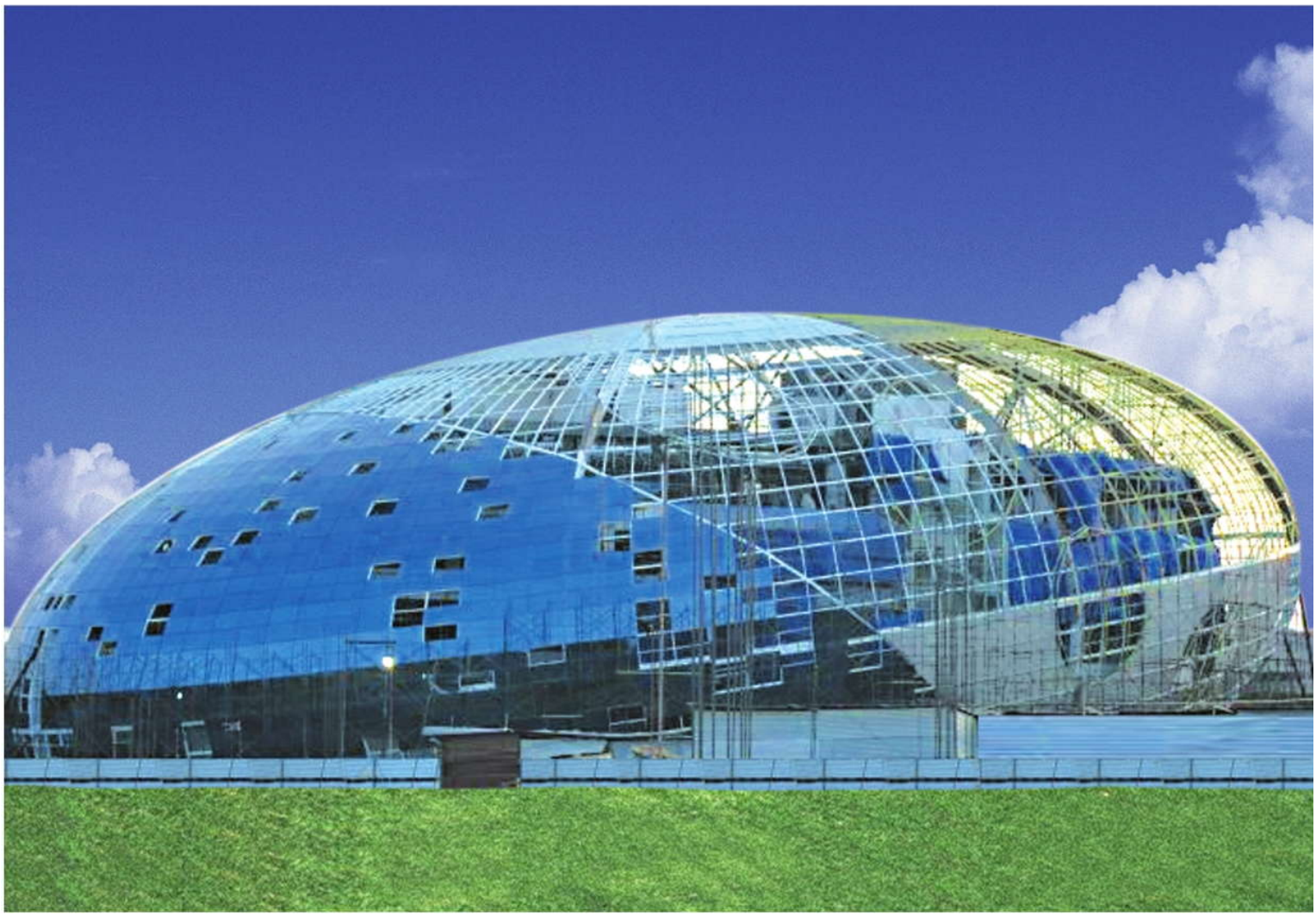
Applications

- Shopping Malls
- Canopies / Atrium
- Glass Curtain Wall Frames
- Partition Frames
- Space Frames
- Guard Rails & Staircases

Attributes

- Lightweight
- Flexible and easy to form shapes
- Smooth surface finish
- Appealing aesthetics
- Contemporary

Atria Mall, Mumbai



Infosys Building, Pune

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Applications

- Airport Terminal Buildings
- Bridges
- Bus Stands
- Sign Supporting Structures
- Pedestrian Walkovers (Footbridge)
- Sports Galleries
- Railways Platforms / Foot Over Bridges

Attributes

- High strength
- More column free space
- Larger & thicker sections
- Ease of fabrication
- Lightweight



Metro Station, Delhi



Aurangabad Airport

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Applications

- Industrial Sheds
- Trusses, Columns and Purlins
- Material Storage Racks
- Mine Roof Support Systems (cogs, props)
- Pallets
- Pipe Racks
- Conveyor Gantries, Trestles
- Drilling Rigs

Attributes

- High strength to weight ratio
- Cost effective
- Ease of fabrication and erection
- Ease of maintenance
- Free from sharp edges



Navneet Publishers, Ahmedabad



Pipe Rack, ITC Haridwar

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Applications

- Automobile Chassis
- Green House Structures
- Truck and Bus Body Members
- Hoarding Structures
- Amusement Park & Playground Equipment
- Exhibition Stalls
- Scaffolding
- Furniture

Attributes

- Lightweight
- Ease of fabrication
- High torsional resistance
- Minimal painting area
- Appealing aesthetics

Green House



Butterfly Park, Bangalore

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PERMISSIBLE AXIAL COMPRESSIVE STRESS (Yst 310 Grade)																											
l/r	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250	300	350
Ac	186	183	178	169	157	143	128	113	99	87	77	67	60	53	47	42	38	34	31	28	26	23	22	20	18	13	10

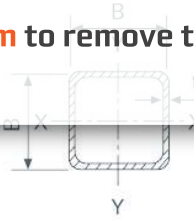
OTHER ALLOWABLE STRESS VALUES (IN Mpa)							
Steel Grade	Minimum Yield Stress	Minimum UTS	Axial Stress in Tension	Bending Stress in Ten. or Compn.	Shear Stress	Bearing Stress	Equivalent Stress
Yst 310	310	450	186	205	140	232	279

Section Type	Grade	Mechanical Properties				Dimensional Tolerance					
		YST	UTS	% of Elongation		Outside dimension	Thickness	Squareness	Corner radius	Weight	
		Mpa	Mpa	<25.4 mm	>25.4 mm					Individual lengths	On lot of 10 MT
RHS/ SHS IS: 4923	YST 210	210	330	12	20	+/-1% with a minimum of +/- 0.50 mm	+/-10%	90 deg. +/- 2 deg.	3t max	10%, -8%	+/- 7.5%
	YST 240	240	410	10	15						
	YST 310	310	450	8	10						
CHS IS: 1161	YST 210	210	330	12	20	OD upto and Incl 48.3 +0.4/-0.8 mm; Over 48.3 mm +/- 1.0%	+ no limit -10%	NA	NA	L +10%, -8%	L +/- 5%
	YST 240	240	410	12	17					M & H +/-10%	M & H +/-7.5%
	YST 310	310	450	12	14						

Length	6.0 m ± 0.05 m Customized length ranging from 6 m to 11 m may be supplied.
Straightness	Minimum 1 : 200th of any length measured along the centre line (mill straightened condition) unless otherwise specifically arranged.
Twist Tolerance	Maximum 2 mm ± 0.5 mm / m length - measured relative vertical shift of any adjacent corner of the section, measured by keeping one side on flat surface.
End Finish	Plain ended - Mechanically sheared, mill - cut finish without further machining.
Surface Finish	Black without any surface treatment of oiling or varnishing.
Raw Material	Sulphur content : 0.05% max, phosphorus content : 0.05% max, equivalent carbon percentage well within specified weldability limits with matching physical properties. For corrosion resistant steel in Cu-bearing variety, refer to the WRS section of this brochure.
Weldability	Tata Structura Steel Hollow Sections are weldable with standard M. S. Electrodes without any pre-heating.
Packing	Bundled by sealing metal strap, and each bundle is labelled for size, measurement, Lot no. etc. Approximate weight of each bundle is 1.5 Mt (+/- 500 kg).
Identification	Marking of **"TATA STRUCTURA"** emblem on surface punched/stenciled/sticker pasted on all Steel Hollow Sections. Standard BIS mark is also put on the sections.
NOTE	Tata Structura Hollow Sections in customized size, grade, length, surface finish and end finish may be delivered as per agreed supply conditions.

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Section Properties



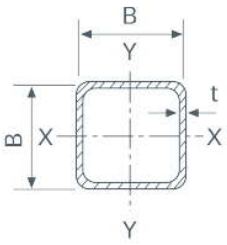
Specification: IS : 4923

Properties of Tata Structura (Square Hollow Sections) YST 310 Grade

SHS B x B mm	Thickness mm	Sec Area A cm^2	Unit W kg/m	Moment of Inertia		Radius of Gyration		Elastic Modulus		Torsional Constants		Outer Surface
				I_{xx} cm^4	I_{yy} cm^4	r_{xx} cm	r_{yy} cm	Z_{xx} cm^3	Z_{yy} cm^3	J cm^4	B cm^3	Area per m m^2
25 x 25	2.00	1.74	1.36	1.48	1.48	0.92	0.92	1.19	1.19	2.29	1.68	0.090
	2.60	2.16	1.69	1.72	1.72	0.89	0.89	1.38	1.38	2.68	1.92	0.087
	3.20	2.53	1.98	1.89	1.89	0.86	0.86	1.51	1.51	2.96	2.07	0.084
32 x 32	2.00	2.30	1.80	3.36	3.36	1.21	1.21	2.10	2.10	5.30	3.05	0.118
	2.60	2.88	2.26	4.02	4.02	1.18	1.18	2.51	2.51	6.45	3.63	0.115
	3.20	3.42	2.69	4.54	4.54	1.15	1.15	2.84	2.84	7.41	4.07	0.112
38 x 38	2.60	3.51	2.75	7.14	7.14	1.43	1.43	3.76	3.76	11.51	5.49	0.139
	3.20	4.19	3.29	8.18	8.18	1.40	1.40	4.30	4.30	13.45	6.28	0.136
	4.00	5.03	3.95	9.26	9.26	1.36	1.36	4.87	4.87	15.67	7.12	0.131
40 x 40	2.60	3.72	2.92	8.45	8.45	1.51	1.51	4.22	4.22	13.63	6.20	0.147
	3.20	4.45	3.49	9.72	9.72	1.48	1.48	4.86	4.86	16.00	7.12	0.144
	4.00	5.35	4.20	11.07	11.07	1.44	1.44	5.54	5.54	18.75	8.12	0.139
50 x 50	2.60	4.76	3.74	17.47	17.47	1.92	1.92	6.99	6.99	28.53	10.37	0.187
	2.90	5.25	4.12	18.99	18.99	1.90	1.90	7.60	7.60	31.15	11.23	0.185
	3.60	6.35	4.98	22.15	22.15	1.87	1.87	8.86	8.86	36.58	12.98	0.181
	4.50	7.67	6.02	25.50	25.50	1.82	1.82	10.20	10.20	41.99	14.68	0.177
60 x 60	2.60	5.80	4.55	31.33	31.33	2.33	2.33	10.44	10.44	50.08	15.52	0.227
	3.20	7.01	5.50	36.94	36.94	2.30	2.30	12.31	12.31	60.02	18.31	0.224
	4.00	8.55	6.71	43.55	43.55	2.26	2.26	14.52	14.52	72.41	21.62	0.219
	4.80	10.01	7.85	49.22	49.22	2.22	2.22	16.41	16.41	83.86	24.51	0.215
72 x 72	3.20	8.54	6.71	66.32	66.32	2.79	2.79	18.42	18.42	106.81	27.47	0.272
	4.00	10.47	8.22	79.03	79.03	2.75	2.75	21.95	21.95	129.85	32.78	0.267
	4.80	12.31	9.66	90.31	90.31	2.71	2.71	25.09	25.09	151.55	37.55	0.263
80 x80	3.20	9.57	7.51	92.71	92.71	3.11	3.11	23.18	23.18	148.55	34.60	0.304
	4.00	11.75	9.22	111.04	111.04	3.07	3.07	27.76	27.76	181.22	41.49	0.299
	4.80	13.85	10.87	127.58	127.58	3.04	3.04	31.89	31.89	212.26	47.77	0.295
91.5 x 91.5	3.60	12.32	9.67	156.49	156.49	3.56	3.56	34.21	34.21	251.17	51.14	0.347
	4.50	15.14	11.88	187.57	187.57	3.52	3.52	41.00	41.00	306.78	61.40	0.343
	5.40	17.85	14.01	215.68	215.68	3.48	3.48	47.14	47.14	359.76	70.77	0.338
100 x 100	4.00	14.95	11.73	226.35	226.35	3.89	3.89	45.27	45.27	364.75	67.50	0.379
	5.00	18.36	14.41	271.10	271.10	3.84	3.84	54.22	54.22	441.84	80.54	0.374
	6.00	21.63	16 .98	311.47	311.47	3.79	3.79	62.29	62.29	511.80	92.06	0.369
113.5 x 113.5	4.80	20.28	15.92	393.30	393.30	4.40	4.40	69.30	69.30	637.45	103.89	0.429
	5.40	22.60	17.74	432.58	432.58	4.38	4.38	76.23	76.23	708.69	114.41	0.426
132 x 132	4.80	23.83	18.71	634.39	634.39	5.16	5.16	96.12	96.12	1018.30	144.11	0.503
	5.40	26.60	20.88	700.11	700.11	5.13	5.13	106.08	106.08	1134.25	159.18	0.500
150 x 150	4.00	22.95	18.01	807.82	807.82	5.93	5.93	107.71	107.71	1273.46	161.38	0.579
	5.00	28.36	22.26	982.12	982.12	5.89	5.89	130.95	130.95	1569.09	196.38	0.574
	6.00	33.63	26.40	1145.91	1145.91	5.84	5.84	152.79	152.79	1856.18	229.44	0.569
	8.00	43.79	34.38	1443.00	1443.00	5.74	5.74	192.40	192.40	2405.78	290.12	0.559
180 x 180	4.00	27.75	21.78	1421.74	1421.74	7.16	7.16	157.97	157.97	2224.31	236.76	0.699
	5.00	34.36	26.97	1736.87	1736.87	7.11	7.11	192.99	192.99	2747.93	289.40	0.694
	6.00	40.83	32.05	2036.52	2036.52	7.06	7.06	226.28	226.28	3259.23	339.65	0.689
	8.00	53.39	41.91	2590.73	2590.73	6.97	6.97	287.86	287.86	4246.16	433.32	0.679
220 x 220	6.00	50.43	39.59	3813.36	3813.36	8.70	8.70	346.67	346.67	6034.53	520.18	0.849
	8.00	66.19	51.96	4894.99	4894.99	8.60	8.60	445.00	445.00	7897.48	668.99	0.839
	10.00	81.43	63.92	5887.19	5887.19	8.50	8.50	535.20	535.20	9549.15	796.48	0.829
	12.00	96.14	75.47	6793.08	6793.08	8.41	8.41	617.55	617.55	11116.96	915.37	0.818
250 x 250	6.0	57.63	45.24	5672.00	5672.00	9.92	9.92	453.76	453.76	8920.44	680.77	0.969
	8.0	75.79	59.50	7315.65	7315.65	9.82	9.82	585.25	585.25	11702.07	879.31	0.959
	10.00	93.43	73.34	8842.29	8842.29	9.73	9.73	707.38	707.38	14248.15	1054.68	0.949
	12.00	110.54	86.77	10254.78	10254.78	9.63	9.63	820.38	820.38	16678.37	1219.59	0.938

* For availability of size/thickness please refer to page 9.

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Specification:
IS : 4923

Product Range: Square Hollow Sections (SHS)

Section Size	Thickness (mm) = t															
	B	B	2.0	2.6	2.9	3.2	3.6	4.0	4.5	4.8	5.0	5.4	6.0	8.0	10.0	12.0
	25	25	■	■		■										
	32	32	■	■		■										
	38	38		■		■		■								
	40	40		■		■		■								
	50	50		■	■		■		■							
	60	60		■		■		■		■						
	72	72				■		■		■						
	80	80				■		■		■						
	91.5	91.5					■		■			■				
	100	100						■			■		■			
	113.5	113.5								■		■				
	132	132								■		■				
	150	150						■			■		■	■		
	180	180						■			■		■	■		
	220	220											■	■	▨	▨
	250	250											■	■	▨	▨

■ - Indicates presently rolled section

▨ - Under development, please confirm availability before adoption in design

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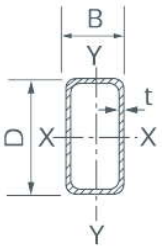
Properties of Tata Structura (Rectangular Hollow Sections)

YST 310 Grade

RHS D x B mm	Thickness mm	Sec Area A cm^2	Unit W kg/m	Moment of Inertia		Radius of Gyration		Elastic Modulus		Torsional Constants		Outer Surface
				I_{xx} cm^4	I_{yy} cm^4	r_{xx} cm	r_{yy} cm	Z_{xx} cm^3	Z_{yy} cm^3	J cm^4	B cm^3	Area per m m^2
50 x 25	2.00	2.74	2.15	8.38	2.81	1.75	1.01	3.35	2.25	6.79	3.79	0.142
	2.60	3.46	2.71	10.16	3.36	1.71	0.99	4.06	2.69	8.27	4.53	0.137
	3.20	4.13	3.24	11.63	3.80	1.68	0.96	4.65	3.04	9.52	5.12	0.134
	4.00	4.95	3.88	13.13	4.23	1.63	0.92	5.25	3.38	10.86	5.69	0.129
60 x 40	2.60	4.76	3.73	22.76	12.09	2.19	1.59	7.59	6.05	25.59	9.83	0.187
	2.90	5.25	4.12	24.74	13.11	2.17	1.58	8.25	6.56	28.02	10.66	0.185
	3.60	6.35	4.98	28.90	15.23	2.13	1.55	9.63	7.62	33.30	12.41	0.181
	4.50	7.67	6.02	33.31	17.44	2.08	1.51	11.10	8.72	39.34	14.29	0.177
66 x 33	2.60	4.70	3.69	25.15	8.43	2.31	1.34	7.62	5.11	20.75	8.71	0.185
	2.90	5.19	4.07	27.33	9.12	2.29	1.33	8.28	5.53	22.65	9.43	0.183
	3.60	6.28	4.93	31.87	10.52	2.25	1.29	9.66	6.37	26.71	10.90	0.179
	4.50	7.58	5.95	36.64	11.93	2.20	1.25	11.10	7.23	31.21	12.43	0.175
80 x 40	2.60	5.80	4.55	46.58	15.74	2.84	1.65	11.65	7.87	38.50	13.46	0.227
	3.20	7.01	5.50	54.94	18.41	2.80	1.62	13.74	9.21	45.83	15.78	0.224
	4.00	8.55	6.71	64.79	21.49	2.75	1.59	16.20	10.74	54.77	18.49	0.219
	4.80	10.01	7.85	73.22	24.03	2.71	1.55	18.30	12.02	62.81	20.79	0.215
96 x 48	3.20	8.54	6.71	98.61	33.28	3.40	1.97	20.54	13.87	82.13	23.82	0.272
	4.00	10.47	8.22	117.54	39.32	3.35	1.94	24.49	16.38	99.11	28.24	0.267
	4.80	12.31	9.66	134.35	44.55	3.30	1.90	27.99	18.56	114.80	32.14	0.263
122 x 61	3.60	12.32	9.67	232.61	78.83	4.34	2.53	38.13	25.84	193.91	44.50	0.347
	4.50	15.14	11.88	278.94	93.78	4.29	2.49	45.73	30.75	235.39	53.13	0.343
	5.40	17.85	14.01	320.83	107.03	4.24	2.45	52.60	35.09	274.29	60.89	0.338
145 x 82	4.80	20.28	15.92	555.16	228.50	5.23	3.36	76.57	55.73	534.27	94.45	0.429
	5.40	22.60	17.74	610.85	250.59	5.20	3.33	84.26	61.12	592.70	103.81	0.426
172 x 92	4.80	23.83	18.71	917.13	346.91	6.20	3.82	106.64	75.41	826.04	128.85	0.503
	5.40	26.59	20.88	1012.47	381.74	6.17	3.79	117.73	82.99	918.10	142.04	0.500
200 x 100	4.00	22.95	18.01	1199.71	410.78	7.23	4.23	119.97	82.16	991.47	141.46	0.579
	5.00	28.36	22.26	1459.25	496.94	7.17	4.19	145.93	99.39	1216.96	171.53	0.574
	6.00	33.63	26.40	1703.31	576.91	7.12	4.14	170.33	115.38	1434.03	199.68	0.569
	8.00	43.79	34.38	2146.21	719.19	7.00	4.05	214.62	143.84	1843.86	250.68	0.559
220 x 140	4.00	27.75	21.78	1892.62	947.66	8.26	5.84	172.06	135.38	2004.80	223.74	0.699
	5.00	34.36	26.97	2313.45	1155.26	8.21	5.80	210.31	165.04	2468.51	272.71	0.694
	6.00	40.83	32.05	2714.10	1351.70	8.15	5.75	246.74	193.10	2914.71	318.93	0.689
	8.00	53.39	41.91	3456.31	1712.25	8.05	5.66	314.21	244.61	3270.10	378.40	0.679
240 x 120	4.00	27.75	21.78	2110.72	725.35	8.72	5.11	175.89	120.89	1736.39	208.03	0.699
	5.00	34.36	26.97	2579.67	882.47	8.67	5.07	214.97	147.08	2138.48	253.55	0.694
	6.00	40.83	32.05	3025.91	1030.45	8.61	5.02	252.16	171.74	2528.39	296.70	0.689
	8.00	53.39	41.91	3851.84	1299.95	8.49	4.93	320.99	216.66	3272.90	376.29	0.679
260 x 180	6.00	50.43	39.59	4855..87	2763.43	9.81	7.40	373.53	307.05	5619.50	501.05	0.849
	8.00	66.19	51.96	6238.69	3538.10	9.71	7.31	479.90	393.12	6821.23	620.13	0.839
	10.00	81.43	63.92	7509.51	4244.26	9.60	7.22	577.65	471.58	8972.14	760.42	0.834
	12.00	96.14	75.47	8672.42	4884.94	9.50	7.13	667.11	542.77	10150.39	875.12	0.829
300 x 150	6.00	51.63	40.53	6073.51	2079.57	10.85	6.35	404.90	277.28	5034.64	478.20	0.869
	8.00	67.79	53.22	7807.95	2654.12	10.73	6.26	520.53	353.88	6559.05	612.64	0.859
	10.00	83.43	65.49	9403.90	3173.71	10.62	6.17	626.93	423.16	8011.67	736.01	0.848
	12.00	98.54	77.35	10866.10	3641.00	10.50	6.08	724.41	485.47	9110.72	829.98	0.838
300 x 200	6.00	57.63	45.24	7370.23	3962.19	11.31	8.29	491.35	396.22	8186.02	650.85	0.969
	8.00	75.79	59.50	9513.66	5097.04	11.20	8.20	634.24	509.70	10722.83	839.51	0.959
	10.00	93.43	73.34	11507.24	6144.30	11.10	8.11	767.15	614.43	13169.70	1015.43	0.948
	12.00	110.54	86.77	13355.84	7107.43	10.99	8.02	890.39	710.43	15215.03	1160.24	0.938

* For availability of size/thickness please refer to page 11.

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Specification:
IS : 4923

Product Range: Rectangular Hollow Sections (RHS)

Section Size			Wall Thickness (mm) = t													
	D	B	2.0	2.6	2.9	3.2	3.6	4.0	4.5	4.8	5.0	5.4	6.0	8.0	10.0	12.0
	50	25	■	■		■		■								
	60	40		■	■		■		■							
	66	33		■	■		■		■							
	80	40		■		■		■		■						
	96	48				■		■		■						
	122	61					■		■			■				
	145	82								■		■				
	172	92								■		■				
	200	100						■			■		■	■		
	220	140						▨			▨		▨	▨		
	240	120						■			■		■	■		
	260	180											▨	▨	▨	▨
	300	150											▨	▨	▨	▨
	300	200											▨	▨	▨	▨

■ - Indicates presently rolled section

▨ - Under development, please confirm availability before adoption in design

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Specification: IS : 1161

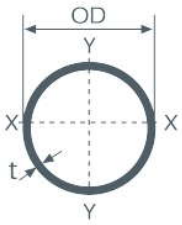
Properties of Tata Structura (Circular Hollow Sections)

YST 310 Grade

Nominal Bore	Outside Diameter	Thickness	Weight	Area of Cross Section	Outer Surface Area/m	Moment of Inertia	Section Modulus	Radius of Gyration
mm	mm	mm	kg/m	cm^2	cm^2/m	cm^4	cm^3	cm
		2.00	0.95	1.21		0.57	0.54	0.69
15	21.3	2.60	1.21	1.53	669	0.69	0.64	0.66
		3.20	1.44	1.82		0.75	0.70	0.65
		2.30	1.38	1.78		1.36	1.01	0.87
20	26.9	2.60	1.56	1.98	845	1.48	1.10	0.86
		3.20	1.87	2.38		1.70	1.26	0.84
		2.60	1.98	2.54		3.09	1.83	1.10
25	33.7	3.20	2.41	3.06	1059	3.61	2.14	1.08
		4.00	2.93	3.73		4.19	2.48	1.05
		2.60	2.54	3.25		6.47	3.05	1.41
32	42.4	3.20	3.01	3.94	1332	7.62	3.59	1.39
		4.00	3.79	4.82		8.99	4.24	1.36
		2.90	3.23	4.13		10.70	4.43	1.61
40	48.3	3.20	3.56	4.53	1517	11.59	4.80	1.59
		4.00	4.37	5.56		13.77	5.70	1.57
		2.90	4.08	5.23		21.59	7.16	2.03
50	60.3	3.60	5.03	6.41	1895	25.88	8.59	2.00
		4.50	6.19	7.88		30.90	10.20	1.98
		3.20	5.71	7.32		48.79	12.82	2.58
65	76.1	3.60	6.42	8.20	2391	54.02	14.20	2.57
		4.50	7.93	10.10		65.12	17.10	2.54
		3.20	6.72	8.61		79.23	17.82	3.03
80	88.9	4.00	8.36	10.70	2793	96.36	21.68	3.00
		4.80	9.90	12.70		112.52	25.31	2.98
		3.60	9.75	12.50		192.03	33.60	3.92
100	114.3	4.50	12.20	15.50	3591	234.30	41.00	3.89
		5.40	14.50	18.50		274.50	48.00	3.85
		4.50	15.00	19.10		437.20	62.60	4.78
125	139.7	4.80	15.90	20.30	4389	463.44	66.35	4.77
		5.40	17.90	22.80		514.50	73.70	4.75
		4.50	17.80	22.70		732.60	88.74	5.68
150	165.1	4.80	18.90	24.20	5187	777.32	94.16	5.67
		5.40	21.30	27.10		864.70	105.00	5.65
		4.80	25.38	32.33		1856.78	169.49	7.58
		6.00	31.51	40.17		2281.95	208.30	7.54
200	219.1	8.00	41.67	53.08	6886	2960.82	270.27	7.47
		10.00	51.59	65.72		3599.89	328.61	7.40
		12.00	61.32	78.11		4201.57	383.53	7.33
		6.00	39.51	50.30		4487.08	328.72	9.44
250	273	8.00	52.30	66.63	8580	5854.07	428.87	9.37
		10.00	64.89	82.66		7156.97	524.32	9.31
		12.00	77.27	98.43		8399.52	615.35	9.24
		6.30	49.36	62.88		7932.09	489.79	11.23
300	323.9	8.00	62.35	79.43	10180	9914.07	612.17	11.17
		10.00	77.44	98.65		12163.24	751.05	11.10
		12.00	92.34	117.63		14325.32	884.55	11.04
		8.00	68.61	87.40		13206.69	742.78	12.29
350	355.6	10.00	85.27	108.62	11176	16230.03	912.83	12.22
		12.00	101.73	129.59		19147.18	1076.89	12.16

* For availability of size/thickness please refer to page 13.

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Specification:
IS : 1161

Product Range: Circular Hollow Sections (CHS)

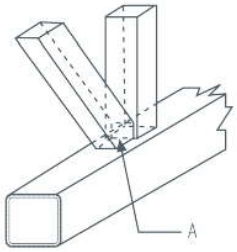
Nominal Bore/Outside Diameter (mm)		Wall Thickness (mm) = t														
NB	OD	2.0	2.3	2.6	2.9	3.2	3.6	4.0	4.5	4.8	5.4	6.0	6.3	8.0	10.0	12.0
15	21.3	■		■		■										
20	26.9		■	■		■										
25	33.7			■		■		■								
32	42.4			■		■		■								
40	48.3				■	■		■								
50	60.3				■		■		■							
65	76.1					■	■		■							
80	88.9					■		■		■						
100	114.3						■		■		■					
125	139.7								■	■	■					
150	165.1								■	■	■					
200	219.1									■	■	■	■	■	▨	▨
250	273									■	■	■	■	■	▨	▨
300	323.9									■	■	■	■	■	▨	▨
350	355.6												▨	▨	▨	▨

■ - Indicates presently rolled section

▨ - Under development, please confirm availability before adoption in design

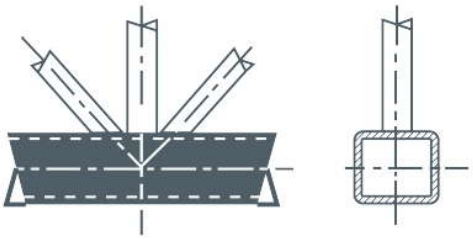
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OVERLAP JOINTS



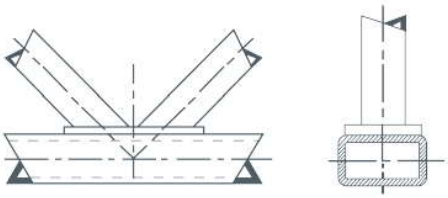
The weld seam 'A' can be omitted without affecting the behaviour of the joint.

GAP JOINTS



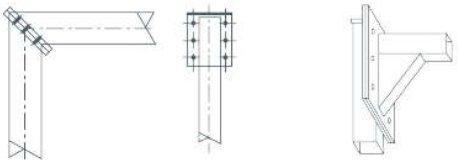
This arrangement is often the simplest and most economical and the joints are sufficiently strong. Joining member alignments should be at > 30^0 with respect to the other.

POSSIBLE REINFORCEMENTS



In certain cases reinforced joints are preferred for improved joint rigidity.

BOLTED KNEE JOINTS



Type A Type B

These are simple and economical arrangements offering good strength and rigidity.

Jointing: Workshop & Site Practice

CUTTING

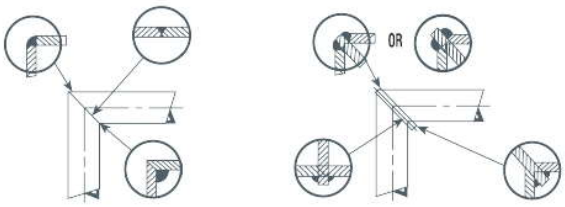
Tata Structura Steel Hollow Sections can be cut:
- By means of a heavy duty circular/hand saw
- By flame cutting: either manual or automatic

- The path of the cut can be marked directly on the surface of the section or on a template after shop layout
- For section thicknesses of 5 mm and above, edges may be chamfered for proper welding penetration

BENDING

- Axial cold bending of Tata Structura Steel Hollow Sections is possible by using an internal mandrel and the roller must be adapted to the shape and size of the section
- Three roll bending machine may be adapted - bend by slow multiple pass, through trial and error method
- Thicker or larger sections are recommended to be pre-heated in a normalising furnace before bending in hot condition for better formation

WELDED KNEE JOINTS



Type A Type B

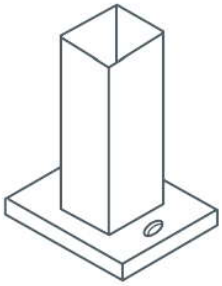
This is simple inexpensive and quite satisfactory, if the connections are not heavily loaded (Type A).
When the sections differ in sizes, or for additional strength, stiffening plates may be used (Type B).

Thickness of stiffening plate 2 x thickness of Tata Structura Rectangular / Square Steel Hollow Sections used or 6 mm, whichever is maximum.

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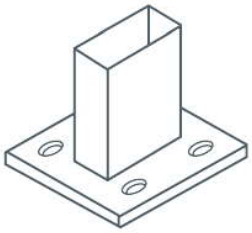
COLUMN BASES

Type A



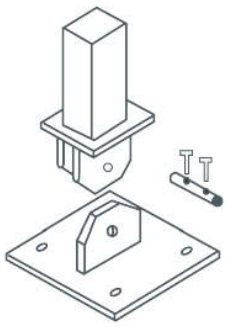
When perfect pin end is not required, (Type A) may be followed and column base with small moment can be made as shown (Type B)

Type B

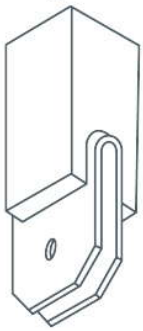


HINGED COLUMN BASES

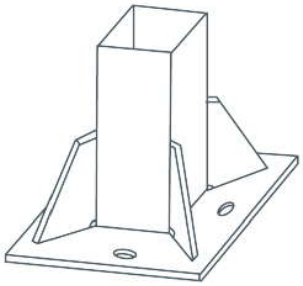
Type A



Type B



Type C



When design assumptions specify an effective hinge in particular plane, this must be implemented by means of an axle or other device imparting rotational freedom as shown (Type A & Type B).

For large axial loads, the colum base should be stiffened as shown (Type C) to minimise the thickness of base plate. Stiffener plates at the middle of Tata Structura Rectangular & Square Steel Hollow Sections' sides better be avoided.

Jointing: Workshop & Site Practice

WELDING

Technique in principle is similar for that of conventional sections. Follow relevant BIS code of practice and design conditions.

- **Electrodes:** Low hydrogen electrodes are suggested for use.
- **Butt welds:** The throat thickness of the seam:
  a) Wall thickness of the section when joining members are of equal thickness. b) Wall thickness of thinner section, if thicknesses are different. Backing strip may be provided to ensure total root penetration in case of thicker section design size.
- **Fillet welds:** Various types may be provided. Size of the fillet is guided by the throat thickness as explained above.
- **NOTE:** All free ends of Tata Structura Steel Hollow Sections should be sealed properly by welding, to prevent internal corrosion.
- Normal M. S. electrodes of reputed brands are recommended. Moisture from electrode should be removed by baking before welding.
- **Sequence:** Edges are to be tack welded to maintain uniform gap during welding to minimise residual stress:

- Transverse weld before longitudinal one
- Fillet weld following butt weld
- Starting from inside to outwards.

TRUSS TO COLUMN CONNECTIONS



Fig. A

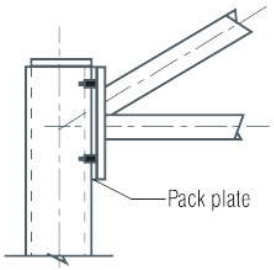


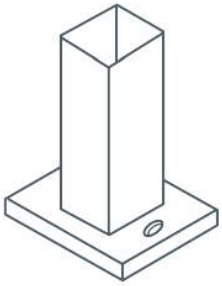
Fig. B

Truss either can rest upon the column as shown (Fig. A) or can be bolted at the face of the column as shown (Fig. B). In the former case the discrepancy in fabrication can be accommodated by providing slotted holes while in the latter the same can be adjusted by packing plates.

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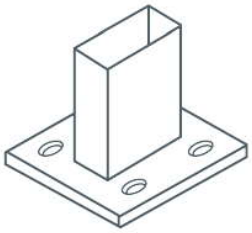
COLUMN BASES

Type A



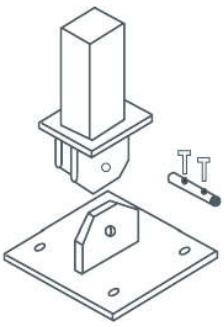
When perfect pin end is not required, (Type A) may be followed and column base with small movement can be made as shown (Type B)

Type B

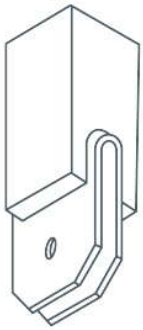


HINGED COLUMN BASES

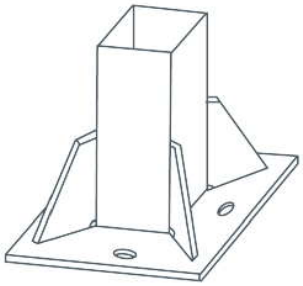
Type A



Type B



Type C



When design assumptions specify an effective hinge in particular plane, this must be implemented by means of an axle or other device imparting rotational freedom as shown (Type A & Type B).

For large axial loads, the colum base should be stiffened as shown (Type C) to minimise the thickness of base plate. Stiffener plates at the middle of Tata Structura Rectangular & Square Steel Hollow Sections' sides better be avoided.

Jointing: Workshop & Site Practice

WELDING

Technique in principle is similar for that of conventional sections. Follow relevant BIS code of practice and design conditions.

- **Electrodes:** Low hydrogen electrodes are suggested for use.
- **Butt welds:** The throat thickness of the seam:
  a) Wall thickness of the section when joining members are of equal thickness. b) Wall thickness of thinner section, if thicknesses are different. Backing strip may be provided to ensure total root penetration in case of thicker section design size.
- **Fillet welds:** Various types may be provided. Size of the fillet is guided by the throat thickness as explained above.
- **NOTE:** All free ends of Tata Structura Steel Hollow Sections should be sealed properly by welding, to prevent internal corrosion.
- Normal M. S. electrodes of reputed brands are recommended. Moisture from electrode should be removed by baking before welding.
- **Sequence:** Edges are to be tack welded to maintain uniform gap during welding to minimise residual stress:
  - Transverse weld before longitudinal one
  - Fillet weld following butt weld
  - Starting from inside to outwards.

TRUSS TO COLUMN CONNECTIONS



Fig. A

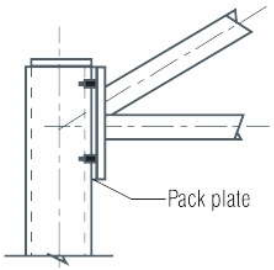


Fig. B

Truss either can rest upon the column as shown (Fig. A) or can be bolted at the face of the column as shown (Fig. B). In the former case the discrepancy in fabrication can be accommodated by providing slotted holes while in the latter the same can be adjusted by packing plates.

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